

Akshay Dahiya

Protocol Engineer with 8+ years building blockchain infrastructure, smart contracts, and distributed systems.

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TECHNICAL SKILLS

Blockchain & Protocol	Solidity, EVM Architecture, Consensus Systems, EIP-712 Relayers, Tokenomics, DeFi, IPFS
Languages & Backend	Go, Python, Node.js, Rust, PostgreSQL, Redis, Docker, AWS, Linux
Web3 Tools	Hardhat, Foundry, Ethers.js, Web3.py
AI & Machine Learning	PyTorch, TensorFlow, Reinforcement Learning (PPO), Computer Vision, Agentic AI
Frontend	React, Next.js, TypeScript, JavaScript

KEY HIGHLIGHTS

- **800 ETH raised** and **5,300+ SBT mints** through protocol smart contracts
- **1M+ gasless transactions** sponsored via EIP-712 relayers
- **50+ production smart contracts** deployed across EVM chains
- **3M+ requests/day** API platform at scale
- **2 ACM publications**; research featured in Forbes
- **2,000+ projects reviewed**, 500+ students mentored

PROFESSIONAL EXPERIENCE

Lead Protocol Engineer

Dec 2022 - Present

POWERLOOM PROTOCOL

Remote

- Architected protocol evolution from V1 to V2 to DSV, designing core consensus and data verification systems
- Developed and deployed Snapshotter node contracts generating **5,300+ SBT mints** and **800 ETH** in protocol revenue
- Designed **\$POWER tokenomics** including dynamic-tax staking, Merkle airdrop infrastructure, and shipped mainnet on Arbitrum Nitro
- Built high-throughput **gasless EIP-712 relayers** with batching and replay protection, sponsoring **1M+ transactions**
- Developed **Onchain Points system** and Generative Prediction Markets with auto-resolution on live protocol data
- Led engineering for **3 incentivized testnets** with strategic partners including CoinList, Polygon zkEVM, and Socket
- Single-handedly shipped **BDS (Block Data Service)** — production decentralized data market streaming real-time Uniswap data with cryptographic IPFS proofs

Senior Software Engineer

Mar 2020 - Dec 2022

HYPERSONIX INC.

Remote

- Built **API Gateway and Management Platform** from scratch, scaling to **3M+ requests/day**
- Improved API performance by **80%** through code optimization, re-architecture, and Brotli compression
- Implemented enterprise authentication using Auth0 (Django Auth, API keys, OAuth 2.0, SSO)
- Designed and deployed ELK-based logging, performance monitoring, and developer learning center

Blockchain Developer

Dec 2020 - Nov 2022

FREELANCE

Remote

- Deployed **50+ production smart contracts** across Ethereum, Avalanche, BSC, and Polygon
- Architected ERC-20, ERC-721, ERC-1155 tokens including hyper-deflationary and dynamic reward mechanisms
- Built DeFi protocols with evolving NFTs, referral systems, and MEV-resistant vaults
- Developed Play-to-Earn gaming systems with custom Chainlink VRF oracles and cross-chain bridges

Research Assistant — Dr. Thad Starner

Jan 2020 - May 2021

GEORGIA INSTITUTE OF TECHNOLOGY

Atlanta, GA

- Built **Examinator** — ML-powered plagiarism detection system analyzing **2M+ exam pairs**; improved detection accuracy from 15% to 25%
- Developed **JackMarker** (traceable document tokens) and **GitDown** (automated leak removal); featured in **Forbes** and **Times Higher Education**
- Published **2 peer-reviewed papers** at ACM Learning@Scale; systems remain in production at Georgia Tech

Graduate Programmer

Jun 2018 - Nov 2018

FIDELITY INTERNATIONAL

Bangalore, India

- Built real-time survey application using WebSockets, Django, Redis, and D3.js visualization
- Developed React-based tools for portfolio managers in the behavioral finance team

Technical Mentor & Project Reviewer

Feb 2017 - Mar 2020

UDACITY

Remote

- Reviewed **2,000+ projects** across Full Stack, Deep Learning, and React nanodegrees
- Mentored **500+ students** through technical challenges, code reviews, and debugging sessions

Open Source Contributor & Admin

Feb 2018 - Aug 2021

GOOGLE SUMMER OF CODE — HYDRA ECOSYSTEM

Remote

- Core contributor to Hydra Ecosystem — Linked Data framework for semantic Web APIs
- Served as GSoC Admin and Mentor (2018–2021), mentoring **10+ contributors**

EDUCATION

Master of Science in Computer Science — Machine Learning Specialization

2019 - 2021

GEORGIA INSTITUTE OF TECHNOLOGY

Atlanta, GA

- Research: Counter-plagiarism systems under Dr. Thad Starner (pioneer in wearable computing)
- Coursework: Machine Learning, Reinforcement Learning, Computer Vision, AI, Graduate Algorithms

Bachelor of Technology in Electronics and Communication Engineering

2014 - 2018

DCRUST

India

- Comprehensive training in electronics, communication systems, and core engineering principles
- Strong foundations in signal processing, embedded systems, and technical problem-solving

Nanodegree Certifications (4)

2016 - 2020

UDACITY

Online

- **Deep Learning (2020):** Neural networks with PyTorch — CNNs, RNNs, LSTMs, GANs
- **Computer Vision (2020, NVIDIA):** YOLO, R-CNN, Visual SLAM, Image Captioning
- **React (2017):** 3 full-stack apps with Redux — MyReads, Would You Rather, Mobile Flashcards
- **Full Stack (2016):** PostgreSQL, RESTful APIs, OAuth, AWS deployment

Open Courseware Foundations

2015 - 2019

MIT OCW / STANFORD / BERKELEY

Online

- **MIT 6.006:** Core algorithms — sorting, heaps, BSTs, graph algorithms, dynamic programming
- **MIT 6.042J:** Discrete math — proofs, graph theory, probability
- **Stanford CS161:** Divide-and-conquer, randomized algorithms, NP-completeness
- **Berkeley CS188:** AI fundamentals — search, MDPs, reinforcement learning, Q-learning

PUBLICATIONS

1. Dahiya, A., Graziano, R., Irish, I., Starner, T.

JackMarker with GitDown: A Framework to Counter Plagiarism at Scale

ACM Learning @ Scale 2021. DOI: 10.1145/3430895.3460982

2. Apoorv, R., Dahiya, A., Graziano, R., Starner, T.

Examinator: A Plagiarism Detection Tool for Take-Home Exams

ACM Learning @ Scale 2020. DOI: 10.1145/3386527.3406723